VIVEK TIWARI

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RESEARCH PROFILE

Eco-hydrological modeller working on agricultural nitrogen and water quality in the Indo-Gangetic Plain. My doctoral research uses coupled surface water and groundwater modelling to quantify how nitrogen applied to intensively irrigated cropland is redistributed between soil, aquifer, and atmosphere over multi-decadal timescales, and to evaluate management practices that reduce groundwater nitrate without compromising crop yields. I work within an India-Netherlands collaborative project and have presented this research at international meetings in Asia and Europe.

Research interests: Eco-hydrological and water quality modelling (SWAT+, MODFLOW); agricultural nitrogen cycling; groundwater nitrate dynamics; greenhouse gas emissions from cropland; best management practice evaluation; remote sensing for water resources

EDUCATION

Ph.D., Water Resources Development and Management, IIT Roorkee — 2022 – present

CGPA 8.88. Thesis: eco-hydrological modelling of agricultural best management practices for water and soil quality improvement in the Hindon River Basin.

M.Tech., National Institute of Technology Raipur — 2019 – 2021

CGPA 7.55

B.E., Chhattisgarh Swami Vivekananda Technical University, Bhilai — 2015 – 2019

CGPA 8.00

RESEARCH EXPERIENCE

Senior Research Fellow, Dept. of Water Resources Development and Management, IIT Roorkee — 2023 – present

- Doctoral research within the HIROS project (Hindon Root Sensing: River Rejuvenation through Scalable Water and Solute Balance Modeling and Informed Farmers Action), an India-Netherlands collaboration under the Clean Ganga Mission.
- Developed a coupled SWAT+ and gwflow model of the Hindon River Basin (4,557 km^2, 725 hydrological response units) to simulate water, nitrogen, and carbon dynamics under rice-wheat and sugarcane rotations.
- Constructed a long-term nitrogen budget across soil, crop, groundwater, and atmospheric compartments, and designed a factorial set of best management practice scenarios combining tillage and nutrient management options.
- Compiled and quality-controlled long-term groundwater nitrate and stream water quality records from national monitoring archives, including censored-data handling and gap filling.
- Built reproducible Python workflows for scenario automation, model output extraction, and multi-criteria evaluation of management alternatives.

Junior Research Fellow, Dept. of Water Resources Development and Management, IIT Roorkee — Mar 2022 – Jul 2023

- Project associate on the HIROS project. Contributed to catchment model setup, land use and management data assembly, and water quality data processing for the Hindon basin.
- Contributed to a parallel modelling study on land use conversion from corn to switchgrass and its implications for water quality and bioenergy production.

Design Engineer, Satyavani Constructions — Aug 2021 – Feb 2022

- Member of the design team for rural water distribution networks across 70 villages under the Jal Jeevan Mission, Government of India.

INTERNATIONAL RESEARCH EXPOSURE

1st mHM Workshop, Helmholtz Centre for Environmental Research (UFZ), Leipzig — 3 – 7 Nov 2025

- Selected participant, with travel and accommodation grant awarded by the organisers. Training in the mesoscale hydrological model (mHM).

Research visit, Wageningen University & Research and TU Delft, Netherlands — 2 – 13 Dec 2024

- Collaborative research stay under the HIROS project, including model intercomparison discussions and presentation of Hindon basin results to Dutch project partners.

PEER-REVIEWED PUBLICATIONS

Tiwari, V.*, & Verma, M. (2023). Prediction of groundwater level using advanced machine learning techniques. *2023 3rd International Conference on Intelligent Technologies (CONIT)*, 1–6. IEEE. doi:10.1109/CONIT59222.2023.10205583

* Corresponding author.

CONFERENCE PRESENTATIONS — ORAL

Tiwari, V., Ilampooranan, I., Vinnarasi, R., & Jain, S. K. (2025). Evaluating non-point source pollution in the Indo-Gangetic Plains: a case study of the Hindon River, India. *22nd Annual Meeting of the Asia Oceania Geosciences Society (AOGS 2025)*, Singapore.

Tiwari, V., Ilampooranan, I., Vinnarasi, R., & Jain, S. K. (2025). Unraveling agricultural non-point source pollution in the Hindon River Basin: a SWAT+ model approach. *XII Scientific Assembly of the International Association of Hydrological Sciences (IAHS)*.

Tiwari, V., Ilampooranan, I., & Vinnarasi, R. (2023, 2024). Eco-hydrological modeling for improving water and soil quality through best agricultural management practices: a Hindon case study. Presented at:

HIROS Learning and Knowledge Sharing Workshop, Delft, Netherlands (2024)
HIROS Convergence Workshop, Indian Institute of Science, Bangalore, India (2024)
International HIROS Partners' Workshop, ICAR-CSSRI, Karnal, India (2023)

Tiwari, V., Sharma, A., & Ilampooranan, I. (2023). Time to switch: switchgrass for water quality improvements and biofuel production. *International SWAT Conference 2023* (online).

Tiwari, V.*, & Verma, M. (2023). Prediction of groundwater level using advanced machine learning techniques. *3rd International Conference on Intelligent Technologies (CONIT)*, Hubli, India.

CONFERENCE PRESENTATIONS — POSTER

Tiwari, V., Ilampooranan, I., Vinnarasi, R., & Jain, S. K. (2026). Can agricultural nitrate leaching to groundwater be reduced without compromising crop yields? *EGU General Assembly 2026*, Vienna, Austria.

Tiwari, V., & Ilampooranan, I. (2025). Eco-hydrological modeling for improving water and soil quality through best agricultural management practices: a Hindon case study. *DST-NWO Cleaning the Ganga Mid-term Meeting*, New Delhi, India, 7 February 2025.

Tiwari, V., Sharma, A., & Ilampooranan, I. (2025). Corn to switchgrass: an eco-hydrological solution for water quality and sustainable energy. *Research Scholar Day*, *Department of Water Resources Development and Management*, *IIT Roorkee*, 15 April 2025.

Tiwari, V., Sharma, A., & Ilampooranan, I. (2023). Corn to switchgrass: an eco-hydrological solution for water quality and sustainable energy. *AGU Fall Meeting 2023*, San Francisco, USA (abstract GC15-06).

GRANTS, AWARDS, AND FELLOWSHIPS

- Travel and accommodation grant, 1st mHM Workshop, Helmholtz Centre for Environmental Research (UFZ), Leipzig, Germany, 2025.
- Senior Research Fellowship, Indian Institute of Technology Roorkee, 2023 – present.
- Qualified Graduate Aptitude Test in Engineering (GATE), 2019 and 2021.

TECHNICAL SKILLS

Hydrological and water quality modelling: SWAT, SWAT+ (including gwflow coupling and source-code level modification), MODFLOW, DSSAT, HEC-HMS, HEC-RAS, TUFLOW, EPANET

Programming and analysis: Python, R, LaTeX, Power BI

Remote sensing and GIS: Google Earth Engine, ArcGIS, QGIS, ERDAS Imagine

Other: AutoCAD, WaterGEMS

TRAINING AND WORKSHOPS

- Hydrological modelling using R, Department of Water Resources Development and Management, IIT Roorkee.
- Short-term course on groundwater hydrogeological investigations, National Institute of Technology Raipur, covering delineation of groundwater potential zones using electrical resistivity analysis.

DEPARTMENTAL SEMINARS PRESENTED

Management control on soil organic carbon; analysis of factors controlling soil organic matter levels in Great Plains grasslands (CENTURY model); utilisation of satellite-based hydrometeorological products in watershed hydrology; rainfall-runoff modelling using ArcSWAT; flood inundation mapping; prediction of groundwater levels using machine learning.

LANGUAGES

Hindi (native), English (professional working proficiency), Chhattisgarhi (native).

References: Available on request.